VAST: Visual Abstractions as Streams of Thought

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Abstract

Visual understanding is as fundamental as language, anchoring how we reason about the physical world. Yet modern vision-language models (VLMs) reason primarily by producing language tokens and rely on pretrained language backbones consuming most parameters and compute. We introduce Visual Abstractions as Streams of Thought (VAST), a cognitively-motivated VLM that reasons through *visual streams of thought*: lower-dimensional abstractions (camera pose, depth, optical flow, point tracks, and text) disentangled from appearance and interpretable when predictions fail. VAST offers two capabilities beyond standard VLMs. First, richly controllable world modeling: conditioning on any combination of modalities (e.g., optical flow, depth, text, point tracks) with patch-level independence for targeted object manipulations, enabling fine-grained counterfactual reasoning. Second, a single model unifies diverse visual-linguistic tasks, e.g., novel view synthesis, captioning, action-conditioned generation, by expressing each as a stream of thought. Together, VAST offers a path toward reasoning that is grounded in the physical world.

Introduction

Visual understanding is critical, anchoring how we reason about the physical world. Infants possess physical knowledge about the visual world, e.g., object permanence, support, containment (Baillargeon et al., 1985; Spelke et al., 1992). Moreover, when people predict physical object motions, they simulate visually (Battaglia et al., 2013; Shepard & Metzler, 1971).

Yet, modern vision-language models (VLMs) reason primarily by producing language tokens, even for questions about motion, forces, and spatial relations. Furthermore, they begin with a pretrained language backbone which consumes the vast majority of parameters and training compute, after which vision is bolted on as an input encoder (Alayrac et al., 2022; Liu et al., 2024).

We introduce VAST (**V**isual **A**bstractions as **S**treams of **T**hought), a cognitively-motivated alternative that reasons through visual abstractions rather than language alone. VAST is a single autoregressive transformer that operates over a unified token vocabulary spanning RGB frames, camera pose, depth, optical flow, point tracks, and text. This design enables two capabilities beyond standard VLMs: (1) richly controllable world modeling: conditioning on any combination of modalities (e.g., optical flow, depth, text, point tracks) with patch-level independence for targeted object manipulations, enabling fine-grained counterfactual reasoning; and (2) unifying diverse visual-linguistic tasks in a single model, e.g., novel view synthesis, captioning, action-conditioned generation, by expressing each as a stream of thought over interpretable visual abstractions.

Methods

Architecture. VAST is a single 8B-parameter autoregressive transformer built on the Probabilistic Structure Integration (PSI) visual world model (Kotar et al., 2025). PSI is a token-based autoregressive transformer trained to jointly model visual structure across several modalities (Kim et al., 2025; Lee et al., 2025): RGB frames (RGB), camera pose (C), depth (D), and optical flow (F), treating each spatial patch as independent tokens and learning distributions over their joint ordering. VAST extends PSI with two additional token types, point tracks (P) and text (T), enabling correspondence-based control and language-grounded reasoning.

Training. We train the model with flexible orderings of abstractions. Training data is organized into several groups: (1) vision-only: RGBCDF sequences on large-scale video datasets, (2) text-only: FineWeb-Edu (deduplicated) (Penedo et al., 2024), (3) vision-text alignment: global and time-indexed dense captions, and (4) points-vision-text alignment: associating point tokens with physical locations in the world and text tokens. The model is trained for 700k steps on vision-only data, followed by 500k steps on a mixture of all groups, with a batch size of 512 and sequence length of 4096.

Richly Controllable World Modeling. Modern VLMs allow only high-level control over visual generation, e.g., text conditioning or discrete commands such as camera control or game-like WASD inputs (Bruce et al., 2024). These models also struggle with maintaining consistency across frames, producing visually appealing yet hallucinated details. In contrast, VAST supports richly controllable world modeling: any combination of modalities (RGB, depth, optical flow, text, point tracks, camera pose) can serve as conditioning input, enabling precise control over spatial, temporal, and semantic aspects of the scene. This builds on PSI's *local patch quantization* and *pointer token* representation. The tokenizer encodes each spatial patch independently, and tags each patch with a pointer token specifying its spatial location. This allows arbitrary spatial ordering during generation and targeted edits to specific scene elements.

Unify Tasks as Streams of Thought. A key property

In submission to the *9th Conference on Cognitive Computational Neuroscience* (CCN 2026).

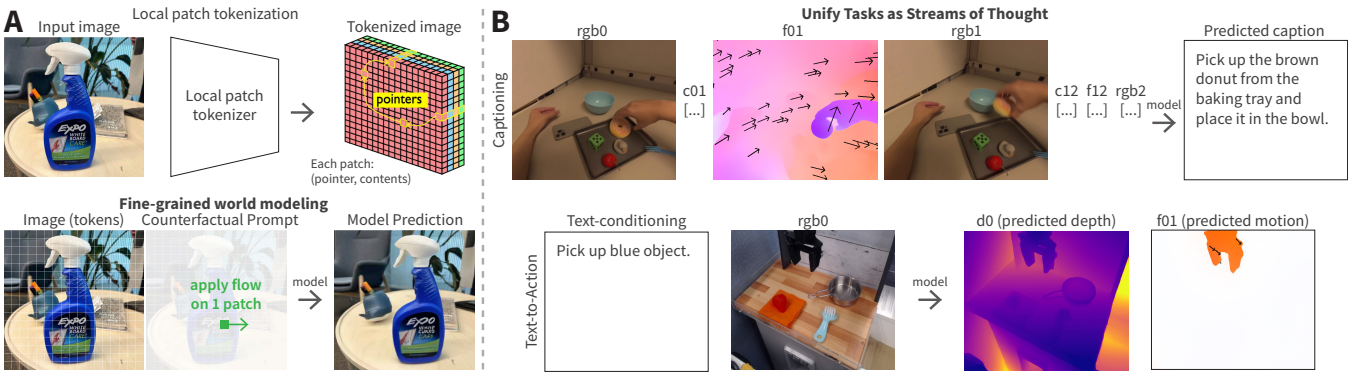


Figure 1: Our model offers two capabilities beyond standard VLMs. (a) Richly controllable world modeling: conditioning on any combination of modalities (e.g., optical flow, depth, text, point tracks), with patch-level independence enabling targeted object manipulations. (b) Unifying diverse visual-linguistic tasks in a single model (e.g., captioning, text-to-action generation) by expressing each task as a stream of thought.

of VAST is that the modality token generation order is not fixed; it is trained over diverse meaningful orderings simultaneously. Because each ordering defines a different task, a single model with a shared vocabulary over all modalities can perform a combinatorial number of tasks without task-specific heads or fine-tuning. Concretely, VAST subsumes many specialist models within a single set of weights: RGB→RGB is video prediction, RGB→depth is monocular depth estimation, RGB_1,RGB_2→flow is optical flow, and RGB,camera→RGB is novel view synthesis. Adding text yields vision-language tasks: text→RGB is generation, RGB→text is captioning, interleaving text with RGB enables time-indexed captioning, and RGB,text→RGB is counterfactual generation. Adding robot joints as points yields robotics tasks: RGB,text→points is language-guided action prediction, RGB,points→RGB is action-conditioned generation, and RGB→points is point tracking. These compose further: RGB,depth,sparse points→dense points,flow,RGB combines tracking, flow estimation, and prediction in a single forward pass.

Results

Richly Controllable World Modeling

VAST enables fine-grained scene manipulation by conditioning on any combination of modalities. For example, specifying motion, through optical flow, to a single patch on an object causes the model to generate consistent visual consequences for the entire scene (Figure 1a).

Unify Tasks as Streams of Thought

These flexible orderings naturally produce *visual streams of thought*: rather than reasoning in language alone, VAST interleaves visual abstractions (depth, optical flow, point tracks) alongside text to perform complex reasoning. Like chain-of-thought (Wei et al., 2023), which externalizes reasoning as a series of language tokens, visual streams of thought externalize reasoning—but through multiple temporally-concurrent representations (depth, flow, point tracks) composable in any order. For example (Figure 1b), given a sequence of interleaved RGB and flow tokens from a manipulation episode, VAST produces a natural-language caption describing the action (e.g., "Pick up the brown donut from the baking tray and place it in the bowl."). Given a text command and an RGB frame, it generates a depth map reasoning about 3D structure and optical flow predicting where the target object should move—yielding grounded action predictions without task-specific heads. These streams are interpretable: if the model errs, we can inspect intermediate abstractions to diagnose whether it misjudged geometry or dynamics.

Moreover, because the model learns general relationships between modalities, it can compose sequences not seen during training. Consider zero-shot segmentation of "the mug on the table": the model first grounds the text query to points on the mug in the scene. It then simulates a small perturbation to those points (as if the mug were nudged slightly to the right) and predicts the resulting optical flow based on this counterfactual. Pixels whose predicted flow is consistent with the perturbation (i.e., co-move with the mug) are assigned to the object; stationary pixels (the table, background) are not. This yields a pixel-level segmentation mask without any segmentation mask supervision during training, emerging purely from the model's learned correspondence between language, point tracks, and optical flow.

The result is a single system that unifies visual-linguistic cognition, in this aspect alike the human brain. An exciting direction is training the model to decide the order of abstractions to predict given a desired task (or discover the visual abstractions themselves), using reinforcement learning or hyperparameter optimization.

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